

Multiple Choice (10 marks)

1. Which ratio can form a proportion with 6 over 9?
 - (a) 8 over 12
 - (b) 9 over 6
 - (c) 2 over 6
 - (d) 24 over 27
2. Find the sum of the primes between 100 and 200, inclusive, that are 1 or 2 more than a perfect square.
 - (a) 298
 - (b) 126
 - (c) 592
 - (d) 396
3. Five thousand dollars compounded annually at an $x\%$ interest rate takes six years to double. At the same interest rate, how many years will it take \$300 to grow to \$9600?
 - (a) 12
 - (b) 1
 - (c) 30
 - (d) 5
4. Which fraction shows the ratio of 3 dogs to 5 dogs?
 - (a) 5 over 3
 - (b) 3 over 8
 - (c) 2 over 5
 - (d) 3 over 5
5. An ant crawls straight from $(-4, 6)$ to the origin, and then continues straight on to $(4, 3)$. How far does it travel?
 - (a) 16
 - (b) $2\sqrt{2}$
 - (c) 5
 - (d) $5+2\sqrt{13}$

True / False (10 marks)

Answer True or False

6. True or False: The mean of the students' test scores is 65.
7. True or False: The total number of T-shirts sold by the volunteers was 643.
8. True or False: The number 6 makes the equation 35 divided by a number equal to 7 true.
9. True or False: When ordered from greatest to least, the numbers are 3, 3.18, 3.8, 3 and 1 over 8.
10. True or False: The sum of the digits of the product of the two smallest 3-digit prime numbers is 8.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. If you continue this pattern in which each line-segment extremity is replaced by a gradually smaller Y in the next figure, in the manner shown, how many endpoints will Figure 5 have? `[asy] draw((0,0)–(0,-3),linewidth(.75)); draw((0,0)–(-2,2),linewidth(.75)); draw((0,0)–(2,2),linewidth(.75)); label("Figure 1",(0,-3),S); draw((5,0)–(5,-2),linewidth(.75)); draw((4,-3)–(5,-2)–(6,-3),linewidth(.75)); draw((4,1)–(5,0)–(6,1),linewidth(.75)); draw((3,1)–(4,1)–(4,2),linewidth(.75)); draw((6,2)–(6,1)–(7,1),linewidth(.75)); label("Figure 2",(5,-3),S); draw((10,0)–(10,-2),linewidth(.75)); draw((9.5,-2.5)–(10,-2)–(10.5,-2.5),linewidth(.75)); draw((9,-2.5)–(9.5,-2.5)–(9.5,-3),linewidth(.75)); draw((11,-2.5)–(10.5,-2.5)–(10.5,-3),linewidth(.75)); draw((9,1)–(10,0)–(11,1),linewidth(.75)); draw((8.5,1)–(9,1)–(9,1.5),linewidth(.75)); draw((11.5,1)–(11,1)–(11,1.5),linewidth(.75)); draw((8.25,.75)–(8.5,1)–(8.25,1.25),linewidth(.75)); draw((8.75,1.75)–(9,1.5)–(9.25,1.75),linewidth(.75)); draw((10.75,1.75)–(11,1.5)–(11.25,1.75),linewidth(.75)); draw((11.75,1.25)–(11.5,1)–(11.75,.75),linewidth(.75)); label("Figure 3",(10,-3),S); [/asy]`
12. Solve the equation $|x - 4| - 10 = 2$, and determine the product of all possible values of x . Explain your reasoning and the steps you took to arrive at your answer.

End of Paper